

UTM TENSILE REPORT — UTM_Test_20260821_171412

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A₀) · anchor self-calibration

42.9 MPa

UTS

35.6 MPa

Yield σ_y

1.55 GPa

Modulus E

9.3 %

Failure ϵ_f

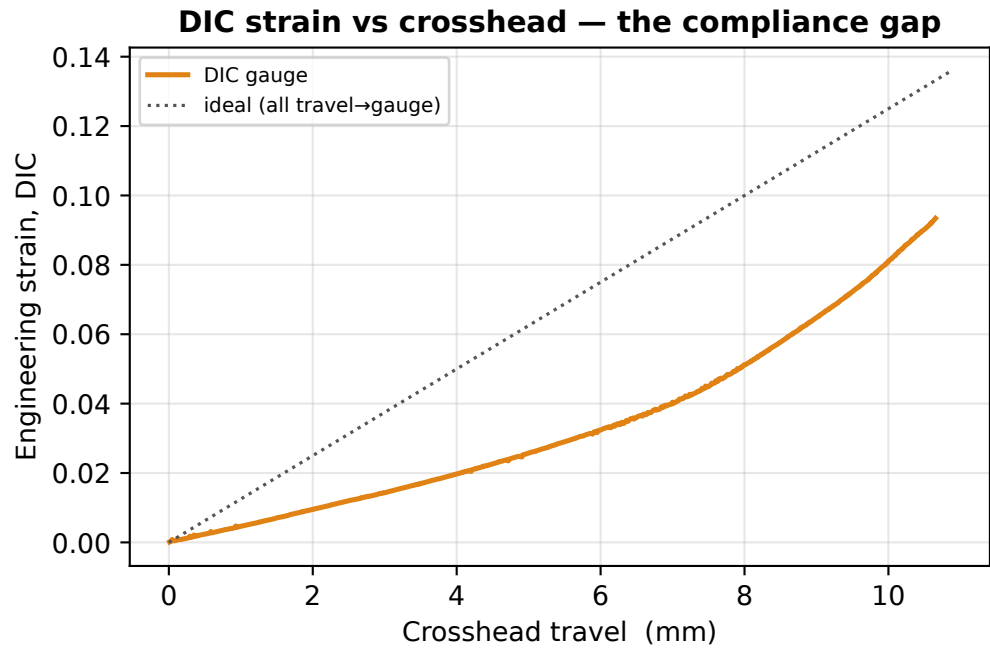
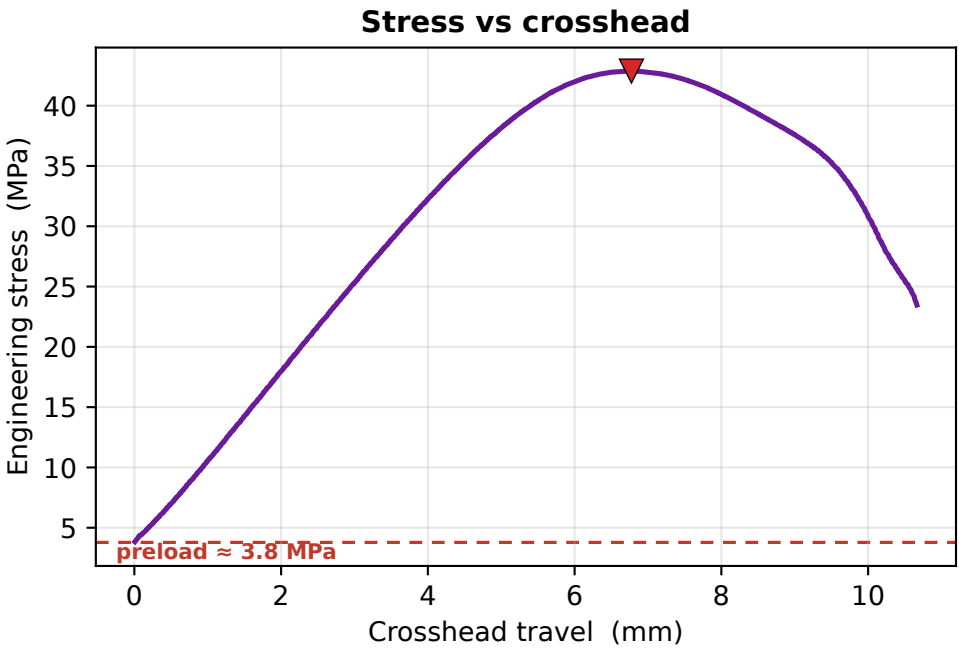
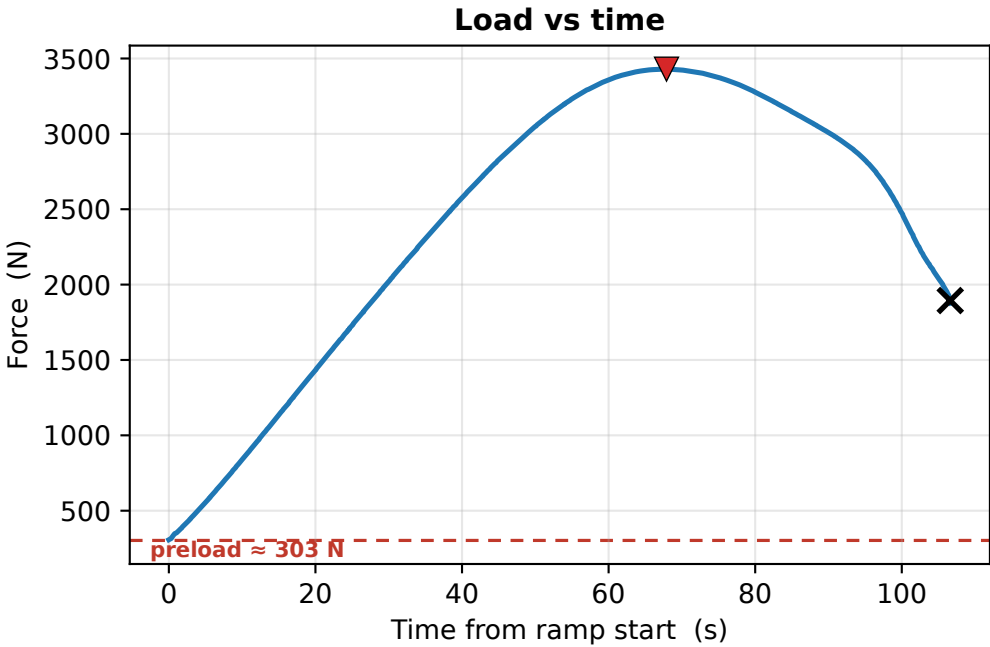
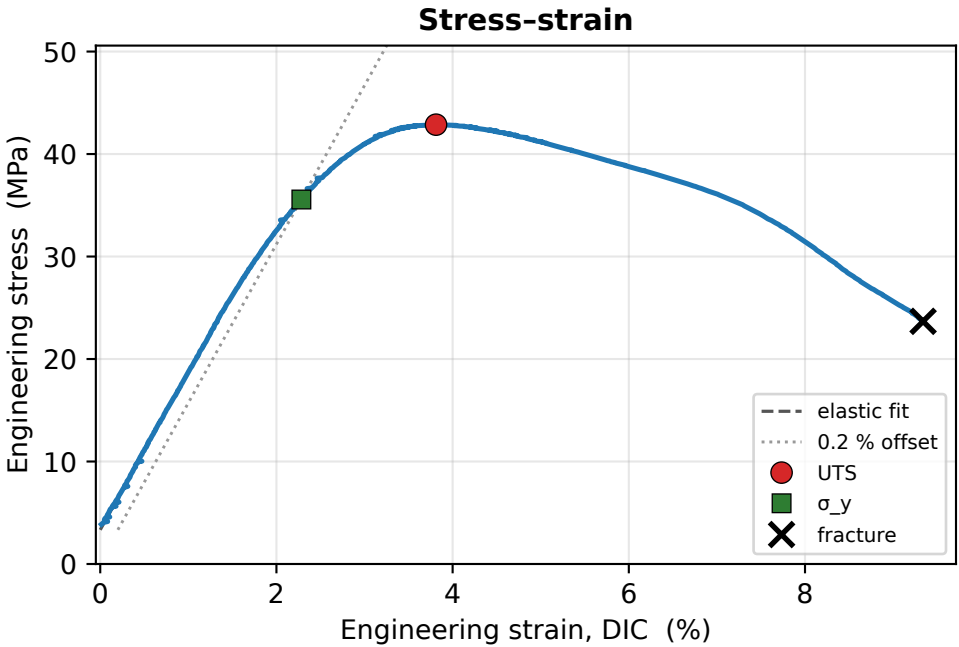
3394 kJ/m³

Toughness

303 N

Preload anchor

TEST SETTINGS	
Specimen	UTM_Test_20260821_171412
DIC mode	Black
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 327.561
Comment	—
Test date	2026-08-21 17:12:01
Duration	138.1 s



VALIDATION vs REFERENCES			
prop	meas	Chacón	E-PLA k
UTS	42.9	32-60 PASS	×1.35
σ_y	35.6	30-50 PASS	×1.63
E	1.5	3-5.5 low	×1.85
ϵ_f	9.3 %	(range 3-7 %)	×0.86
✓ strength inside journal range (k≈1)			